

FILIP RUZICKA

E: ruzicka.filip@gmail.com; W: [website](#), [github](#), [google scholar](#)

EMPLOYMENT

- 2024– IST BRIDGE Fellow (Marie Skłodowska-Curie Actions), **Institute of Science and Technology Austria (Austria)**, with Prof. Beatriz Vicoso
- 2019–23 Postdoc, **Monash University (Australia)**, with A/Prof. Tim Connallon

EDUCATION

- 2015–18 PhD, Evolutionary Genetics, **University College London (UK)**, with Profs. Max Reuter and Kevin Fowler
- 2012–13 MRes Biodiversity, Evolution & Conservation (Institute of Zoology & Natural History Museum, London, UK)
- 2009–12 Bsc Biological Sciences (University College London, UK)

PUBLICATIONS

- 2026 **Ruzicka, F.**, Vicoso, B. (2026). Is faster-X adaptation due to large-effect mutations? An empirical test of a new theory. *Genome Biology & Evolution* (in press) & *bioRxiv*
- Mackintosh, C., Connallon, T., **Ruzicka, F.** (2026). The evolution of context-specific dominance during selective sweeps. *Journal of Evolutionary Biology* (in press) & *bioRxiv*
- Ruzicka, F.** (2026). Reverse genetics of sexual antagonism. *Nature Ecology & Evolution (News & Views)*, 10:1035–1036
- 2025 **Ruzicka, F.***, Zwoinska, M.K.* , Goedert, D., Kokko, H., Li Richter X-Y., Moodie, I.R., Nilén, S., Olito, C., Svensson, E.I., Czuppon, P., Connallon, T. (2025). A century of theories of balancing selection. *Biological Reviews*, 101(2):804-825
- Connallon, T., Czuppon, P., Olito, C., Goedert, D., Kokko, H., Nava-Bolaños, A., Nilén, S., Svensson, E.I., Zwoinska, M.K., Dutoit, L.* , **Ruzicka, F.***. (2025). Predicting the prevalence of genetic trade-offs among adaptive substitutions. *Evolution*, 24:qpaf061
- 2024 McDonough, Y., **Ruzicka, F.**, Connallon, T. (2024). Reconciling theories of dominance with the relative rates of adaptive substitution on sex chromosomes and autosomes. *Proceedings of the National Academy of Sciences U.S.A.*, 121(44): e2406335121
- 2022 **Ruzicka, F.**, Holman, L., Connallon, T. (2022). Polygenic signals of sex differences in selection in humans from the UK Biobank. *PLOS Biology*, 20:e3001768
- Ruzicka, F.**, Reuter, M. (2022). Evolutionary Genetics: Dissecting a sexually antagonistic polymorphism. *Current Biology (Dispatch)*, 35:PR828-R830
- Connallon, T., Beasley, I.J., McDonough, Y., **Ruzicka, F.** (2022). How much does the unguarded X contribute to sex differences in lifespan? *Evolution Letters*, 6:319–329
- Ruzicka, F.**, Connallon, T. (2022). An unbiased test reveals no enrichment of sexually antagonistic polymorphisms on the human X chromosome. *Proceedings of the Royal Society B: Biological Sciences*, 289:202123142
- 2021 **Ruzicka, F.**, Connallon T., Reuter, M. (2021). Sex differences in deleterious mutational effects in *Drosophila melanogaster*: combining quantitative and population genetic insights. *Genetics*, 219:iyab143
- Jardine, M., **Ruzicka, F.**, Diffley, C., Fowler, K., Reuter, M. (2021). A non-coding indel polymorphism in the *fruitless* gene of *Drosophila melanogaster* exhibits antagonistically pleiotropic effects. *Proceedings of the Royal Society B: Biological Sciences*, 288:20202958

2020	Ruzicka, F.* , Connallon, T.*. (2020). Is the X chromosome a hot spot for sexually antagonistic polymorphisms? Biases in current empirical tests of classical theory. <i>Proceedings of the Royal Society B: Biological Sciences</i> , 287:20201869
	Ruzicka, F.* , Dutoit, L.* , Czuppon, P., Jordan, C.Y., Li, X-Y, Olito, C., Runemark, A., Svensson, E.I., Yazdi, H.P., Connallon, T. (2020). The search for sexually antagonistic genes: Practical insights from studies of local adaptation and statistical genomics. <i>Evolution Letters</i> , 4:398-415
2019	Ruzicka, F.* , Hill, M.S.* , Pennell, T.M.* , Flis, I., Ingleby, F.C., Mott, R., Fowler, K., Morrow, E.H., Reuter, M. (2019). Genome-wide sexually antagonistic variants reveal long-standing constraints on sexual dimorphism in fruit flies. <i>PLOS Biology</i> , 17:e3000244
2018	Connallon, T., Olito, C., Dutoit, L., Papoli, H., Ruzicka, F. , Yong, L. (2018). Local adaptation and the evolution of inversions on sex chromosomes and autosomes. <i>Philosophical Transactions of the Royal Society B: Biological Sciences</i> , 373:20170423
2017	Reuter, M., Camus, M. F., Hill, M.S., Ruzicka, F. , Fowler, K. (2017). Evolving plastic responses to external and genetic environments. <i>Trends in Genetics (Forum)</i> , 33:169-170
2016	Davies, T., Ruzicka, F. , Lavery, T., Walters, C.L., Pettoirelli, N. (2016). Ultrasonic monitoring to assess the impacts of forest conversion on Solomon Island bats. <i>Remote Sensing in Ecology and Conservation</i> , 2:107-118
2015	Andujar, C., Arribas, P., Ruzicka, F. , Crampton-Platt, A., Timmermans, M.J.T.N., Vogler, A.P. (2015). Phylogenetic community ecology of soil biodiversity using mitochondrial metagenomics. <i>Molecular Ecology</i> , 24:3603-3617
Manuscripts (available on request)	Ruzicka, F. , Cyza, J., Connallon, T. A genome-wide negative correlation between the severity and dominance of human deleterious mutations.

*denotes shared authorship

FUNDING

2024–26	€134,260, IST BRIDGE Fellowship (Horizon Europe, Marie-Skłodowska Curie Actions, #101034413)
2019	A\$5,000, ECR “Evolution Theme” Leader (Monash School of Biological Sciences) A\$10,000, ECR Career Development Award (Monash School of Biological Sciences)
2014–18	£71,500, PhD Studentship (UK Natural Environment Research Council)

AWARDS

2020	ECR Award (Australasian Evolution Society), including A\$1,000 funding
2018	Runner-up, ECR Talk, the UK Genetics Society’s Autumn Symposium “Genotype to Phenotype to Fitness” Winner, Final Year Talk, UCL GEE Postgraduate Symposium
2017	Winner, Second Year Poster, UCL GEE Postgraduate Symposium
2013	“Darwin Prize” for Top Student, UCL MRes programme

LEADERSHIP

2024–	Co-organiser, “EvoLunch” Seminar Series (ISTA)
2022–23	Co-organiser, “Genetics and Evolution of Sex Differences” symposium (Society for Molecular Biology & Evolution)
2020–21	Co-organiser, School of Biological Sciences Seminar Series (Monash)
2019–21	ECR representative, School of Biological Sciences “Evolution” Theme (Monash)
2016–18	Co-organiser, GEE Friday Seminars for Young Researchers (UCL)
2016	Co-organiser, London Centre for Ecology & Evolution Summer Symposium: “War & Peace: The Dynamics of Evolutionary Conflict” (London), including £3,000 in funding from London Centre for Ecology & Evolution and the UK Genetics Society Filip Ruzicka (ruzicka.filip@gmail.com) — Application for Lecturer in Bioinformatics

TEACHING

2026	Guest lecturer, “Classics in Evolutionary Biology” PhD course (ISTA)
2024–25	PhD tutor, “Population genetics: the basics” PhD course (ISTA)
2020–21	Co-founder, “Foundations of Evolutionary Biology” PhD course in population and quantitative genetics (Monash)
2019–21	Guest lecturer, Evolutionary & Ecological Genetics (Monash), Genomics & Population Genetics (Monash)
2015–21	Undergraduate tutor, Genomics & Population Genetics (Monash), Evolutionary & Ecological Genetics (Monash), Sex Genes & Evolution (UCL), Computational Biology (UCL), Evolutionary Genetics (UCL)

SUPERVISION

2021–22	Naarah Bovingdon (Honours, Monash) Yasmine McDonough (Honours, Monash) Natalie Schmiedl (Undergraduate, Monash)
2020	Justyna Cyza (Undergraduate, Monash)
2019	Isobel Beasley (Undergraduate, Monash)
2018	Olivia Davidson (MRes, UCL)
2016	Harvi Pawar (Undergraduate, UCL) Didem Snaith (Undergraduate, UCL) Max Houghton (MRes, UCL)
2015	Sadé Bates (Msci, UCL)

PRESENTATIONS

2026	Poster, Society for Molecular Biology & Evolution (Copenhagen, Denmark) Talk, 59 th Population Genetics Group (Lille, France)
2025	Talk, EvolVienna (Vienna, Austria) Talk, 58 th Population Genetics Group (Sheffield, UK)
2024	Invited talk, International Laboratory for Human Genome Research, National Autonomous University of Mexico (online) Talk, 3 rd Joint Congress for Evolutionary Biology (Montréal, Canada) Talk, UCL GEE Joint General Meeting (London, UK) Poster, Probabilistic Modelling in Genomics (Vienna, Austria)
2023	Invited talk, Edinburgh University (Edinburgh, UK) Invited talk, Lund University (Lund, Sweden) Invited talk, Institute of Science and Technology Austria (Vienna, Austria) Poster, Jacques Monod Conference “Sex unfolded” (Roscoff, France) Talk, 56 th Population Genetics Group (London, UK)
2021	Talk, Australasian Evolution Society (Melbourne, Australia)
2020	Plenary talk, Australasian Evolution Society (online) Talk, Monash Genetics Genomics & Health seminar series (Melbourne, Australia)
2019	Talk, Australasian Evolution Society (Sydney, Australia) Talk, Monash ECR Science Symposium (Melbourne, Australia) Talk, Monash School of Biological Sciences Seminar Series (Melbourne, Australia) Poster, 27 th European Society for Evolutionary Biology (Turku, Finland) Talk, Genetics Society of Australasia (Melbourne, Australia) Talk, 52 nd Population Genetics Group (Oxford, UK)
2018	Talk, The Genetics Society’s Autumn Symposium “Genotype to Phenotype to Fitness” (Exeter, UK) Talk, UCL GEE Postgraduate Symposium (London, UK) Talk, 51 st Population Genetics Group (Bristol, UK)
2017	Talk, 26 th European Society for Evolutionary Biology (Groningen, Netherlands), including £800 funding to attend (UCL Graduate School Fund) Poster, UCL GEE Postgraduate Symposium (London, UK)
2015	Talk, European Meeting of PhD Students in Evolutionary Biology (Stirling, UK)

Filip Ruzicka (ruzicka.filip@gmail.com) — Application for Lecturer in Bioinformatics

Poster, Quantitative Genomics (London, UK)
Talk, UCL GEE Postgraduate Symposium (London, UK)
Talk, NERC DTP Panel (UCL)

REVIEWING

Peer-reviewed journals *PNAS, Nature Ecology and Evolution, PLOS Biology, PLOS Genetics, Molecular Biology and Evolution, Genome Biology and Evolution, Molecular Ecology, The American Naturalist, Proceedings B, Evolution Letters, Genetics, Evolution, Journal of Evolutionary Biology, Behavioural Ecology and Sociobiology, Heredity*

Funding bodies National Science Foundation (NSF) of the USA, Human Frontiers Science Program
Postdoctoral (HFSP) Fellowships, The Austrian Academy of Sciences DOC fellowships

PUBLIC ENGAGEMENT

2020 Writer, *Evolution Letters* research blog
2018 Scientific expert, 'Agonism/Antagonism' art exhibition at UCL Grant Museum of Zoology
2016–17 Writer, *The Genetics Society* Summer newsletter
2014–17 Tutor, "Generating Genius" volunteer programme aimed at STEM secondary school students from low SES backgrounds
2009–13 Leader, "Roots & Shoots" volunteer programme to communicate scientific and environmental issues to primary school children

TRAINING & WORKSHOPS

2017–24 ESEB Special Topics Network (Lund, Sweden; Montpellier, France; Ingelheim am Rhein, Germany)
2019–20 GENETICS Peer Review Training Programme (online)
2015 "Evolutionary Biology in Guarda" Summer School (Guarda, Switzerland), including £450 funds (University of Basel)

REFEREES

Beatriz Vicoso Professor, Institute of Science and Technology Austria (Vienna, Austria). **E:** bvicoso@ista.ac.at

Tim Connallon A/Professor, School of Biological Sciences, Monash University (Melbourne, Australia). **E:** Tim.connallon@monash.edu

Max Reuter Professor, Dept. of Genetics Evolution & Environment, University College London (London, UK). **E:** m.reuter@ucl.ac.uk